

ECN 711
Problem Set #3

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Due date: September 21st, 2025.

1. Consider the overlapping generations setup with 2-period lives and a perishable consumption good. Preferences are represented by

$$u(c_t^h(t), c_t^h(t+1)) = \log(c_t^h(t)) + \beta \log(c_t^h(t+1))$$

Endowments are given by $w_t^h(t) = \omega_1 h$ and $w_t^h(t+1) = \omega_2 h$. There are finitely many individuals, $h = 1, 2, \dots, N$.

1. State and solve the individual problem.
 2. Define a competitive equilibrium for this economy.
 3. Solve for the equilibrium interest rates. Do equilibrium allocations and interest rates depend on the size of a cohort, N ?
 4. Is there trade in equilibrium? Explain.
2. Consider the overlapping generations setup with 2-period lives and a perishable consumption good. Individuals are identical within a cohort. Preferences are represented by

$$u_t(c_t(t), c_t(t+1)) = \sqrt{c_t(t)} + \beta_t \sqrt{c_t(t+1)}.$$

Suppose that the discount factor of an individual from a cohort born at t , β_t , obeys $\beta_t = \beta_L$ if t is odd, and $\beta_t = \beta_H$ if t is even, with $\beta_L < \beta_H$.

Assume that the cohort size is constant, and that individuals have the same endowment of the consumption good when young and when old.

1. Solve for equilibrium interest rates for all t . How interest rates at different dates compare? Explain.
2. Suppose that individual endowments grow over time: individual endowments from young to old age grow at the constant rate $(1 + g)$. Solve for equilibrium interest rates for all t , and compare your answer to the previous item.
3. Consider the overlapping generations setup with 2-period lives and a perishable consumption good. Preferences are represented by

$$u^h(c_t^h(t), c_t^h(t+1)) = \log(c_t^h(t)) + \beta_h \log(c_t^h(t+1)), \quad h = 1, 2, \dots, N$$

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Endowments are given by $w_t^h(t) = \bar{\omega}$ and $w_t^h(t+1) = \bar{\omega}$, all h .

In words, individuals are heterogeneous in terms of their preferences given that their discount factors differ. We assume that discount factors are ordered according to the individual index, h :

$$\beta_1 < \beta_2 < \beta_3 < \dots < \beta_N$$

1. Solve for the equilibrium interest rate.
2. Show that there is trade in equilibrium. Show that there is an equilibrium threshold value $\hat{\beta}$, so that if $\beta_h < \hat{\beta}$, the individual is a borrower, and if $\beta_h > \hat{\beta}$, the individual is a lender.
4. Consider the overlapping generations setup with a perishable consumption good. Unlike the case discussed in class, we assume that individuals live for three periods. The size of each cohort is constant and equal to N . Preferences are represented by

$$u^h(c_t^h(t), c_t^h(t+1), c_t^h(t+2)) = v(c_t^h(t)) + \beta v(c_t^h(t+1)) + \beta^2 v(c_t^h(t+2))$$

where v is continuous, strictly increasing, differentiable and concave, and $\beta > 0$

1. State the problem of an individual. Obtain the first-order conditions that characterize optimal choices.
2. Define carefully a competitive equilibrium.
3. Suppose now that $u(c) = \log(c)$. Argue that the growth in individual consumption from age 1 to 2, and from age 2 to 3, is the same for all individuals, regardless of the distribution of endowments. Explain.
4. Suppose in addition, that all individuals have the same endowment in every age, $\omega_t^h(t) = \omega_t^h(t+1) = \omega_t^h(t+2) = \bar{\omega}$, all $h = 1, \dots, N$, all t . Solve for the equilibrium interest rate.